Program : **Diploma in Fire Technology and Safety**	
Course Code: **5463A**	Course Title: **Safety in Construction and Textile Industry**
Semester: **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To understand the safety issues unique to the construction and textile industries.
- To identify and mitigate ergonomic risks, equipment hazards, and environmental risks within these industries.
- To introduce relevant regulations, standards, and legal frameworks for ensuring safety in construction.
- To equip students with knowledge about specific safety practices for various construction operations and textile industry hazards.
- To understand the roles of stakeholders in promoting safety and preventing accidents in construction and textile industries.

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand and analyze safety issues in construction and textile industries.	13	Understanding
CO2	Identify and apply safety measures for various construction operations and materials.	16	Applying
CO3	Analyze hazards in construction sites and textile manufacturing units and suggest control measures.	15	Analyzing
CO4	Understand relevant Indian standards, legal frameworks, and welfare provisions for construction workers.	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2				
CO2	3	3	3				
CO3	3	3	2	3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	**Understand and analyze safety issues in construction and textile industries.**		
M1.01	Introduction to the construction industry and its safety issues.	2	Understanding

M1.02	Human factors in construction safety management.	2	Understanding
M1.03	Roles of stakeholders (management, workers, government, contractors) in ensuring safety.	2	Applying
M1.04	Framing of contract conditions on safety and safety-related matters.	2	Understanding
M1.05	Ergonomic risks in construction – identification and control methods.	2	Analyzing
M1.06	Safety in the textile industry – concepts, risks, and safety challenges.	3	Understanding

Contents: **Introduction to Construction and Textile Industry Safety Issues**

Construction Safety Issues: Overview of construction hazards and safety regulations., Identifying and addressing common construction-related risks such as falls, machinery accidents, electrical hazards, and exposure to hazardous materials.
Human Factors in Safety Management: Recognizing human errors and minimizing risks through training, awareness, and design of safe systems, Understanding psychological aspects and behaviors that affect safety.
Stakeholders' Role: Understanding how each participant (management, workers, contractors, government) contributes to safety in construction, Roles of supervisors, safety officers, and unions in promoting a safety culture.
Ergonomics in Construction: Identifying ergonomic risks such as improper lifting techniques, awkward postures, and improper tool design, Suggesting corrective actions like the use of assistive devices, ergonomic tools, and promoting worker training.
Safety in the Textile Industry: Concepts and Significance: Introduction to textile industry operations and common safety challenges. Overview of textile manufacturing processes (spinning, weaving, dyeing, finishing) and related hazards. Common Hazards in the Textile Industry: Mechanical Hazards: Moving parts in machines, risk of entanglement, and injuries related to textile machinery like looms and spinning frames. Fire Hazards: Flammable fibers, lint accumulation, and the risk of fire and explosions. Chemical Hazards: Exposure to hazardous chemicals like dyes, solvents, and finishing agents, leading to skin, respiratory, and eye problems. Ergonomic Risks: Repetitive strain injuries from tasks such as sewing, weaving, or operating machinery for long periods.
Safety Challenges in Textile Industry: Addressing the risks associated with machine guarding, fire prevention, and chemical exposure. Implementing engineering controls, personal protective equipment (PPE), and good housekeeping practices. Training workers on safe operating procedures, first aid, and emergency protocols in case of chemical spills, fires, or accidents.

CO2	**Identify and apply safety measures for various construction operations and materials.**		
M2.01	Safety in the construction of building elements – foundations, excavation, and piling.	4	Applying

M2.02	Safety during the construction of walls, roofs, and structural frames.	3	Applying
M2.03	Safety during the erection of structural steel work and concrete framed structures.	**3**	Applying
M2.04	Safety in construction of temporary structures, including scaffolding and shoring.	**3**	Applying
M2.05	Safety in specialized construction operations – tunneling, blasting, and confined space work.	**3**	Analyzing
	Series Test I	1	

Contents: Safety in Construction Operations and Elements

Construction Phases: Detailed safety measures for foundation work, wall and roof construction, and structural work.
Temporary Structures: Safety standards for scaffolds, ladders, and formwork used during construction.
Specialized Operations: Risk control strategies for tasks such as tunneling, blasting, and confined space work.
Indian Standards and National Building Code: Familiarization with relevant codes and standards for construction safety.

CO3	**Analyze hazards in construction sites and textile manufacturing units and suggest control measures**		
M3.01	Safety in storage, stacking, and handling of construction materials.	3	Applying
M3.02	Safety in the use of construction equipment – hoists, lifts, and cranes.	3	Applying
M3.03	Hazards in construction sites – electrical, fire, and chemical hazards.	3	Analyzing
M3.04	Health hazards at construction sites – physical, psychological, and biological.	3	Analyzing
M3.05	Safety in construction vehicle/machinery operations – excavation, grading, and concrete mixers.	3	Applying

Contents: Safety in Materials Handling and Equipment Material Handling: Safety measures in storing and handling construction materials like cement, aggregates, steel, and hazardous materials. Construction Equipment: Safe operation and maintenance of hoists, cranes, pneumatic tools, and power tools. Construction Hazards: Identifying and controlling electrical, fire, and chemical hazards at construction sites. Health Hazards: Addressing physical, biological, and psychological health risks at construction sites.			
CO4	**Understand relevant Indian standards, legal frameworks, and welfare provisions for construction workers.**		
M 4.01	Requirements for construction workers' habitat and welfare.	2	Understanding
M 4.02	Site selection, minimum facilities, and sanitary provisions.	3	Understanding
M 4.03	The Contract Labour (R&A) Act and Central Rules – Registration, licensing, welfare.	3	Applying
M4.04	The BOCW Act and welfare provisions – Penalties, wages, and legal protections.	3	Understanding
M4.05	Legal frameworks regarding construction safety – overview and implementation.	3	Understanding
	Series Test II	1	
Contents: Welfare Provisions and Legal Framework for Construction Workers Worker Welfare: The minimum facilities required for construction workers, including sanitation, drinking water, and first aid. Legal Framework: An introduction to the Contract Labour (R&A) Act, the BOCW Act, and provisions for worker welfare. Safety Standards: Reviewing the legal safety standards, penalties, and compliance for construction projects. Welfare Regulations: Discussion of health and welfare provisions for workers at construction sites.			

Text /Reference:

T/R	Book Title/Author
T1	Phil Hughes and Ed Ferrett, Introduction to Health and Safety at Work (2nd edn.), Butterworth-Heinemann, UK. (2007)

T2	Helen Lingard and Steve Rowlinson, Occupational Health and Safety in Construction Project Management, Spon Press, Taylor & Francis Group, London. (2005)
R1	K.N. Vaid, Construction Safety Management, National Institute of Construction Management and Research, Mumbai. (1988)
R2	V.J. Davies and K. Tomasin, Construction Safety Handbook, Thomas Telford Publishing, London. (1996)
R3	The Contract Labour (Regulation and Abolition) Central Rules (1971)
R4	Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996 and Central Rules
R5	Relevant Indian Standard Codes of Practice